The American University of Beirut

Faculty of Engineering & Architecture

Sound Visualization for the Deaf

Final Year Project Report

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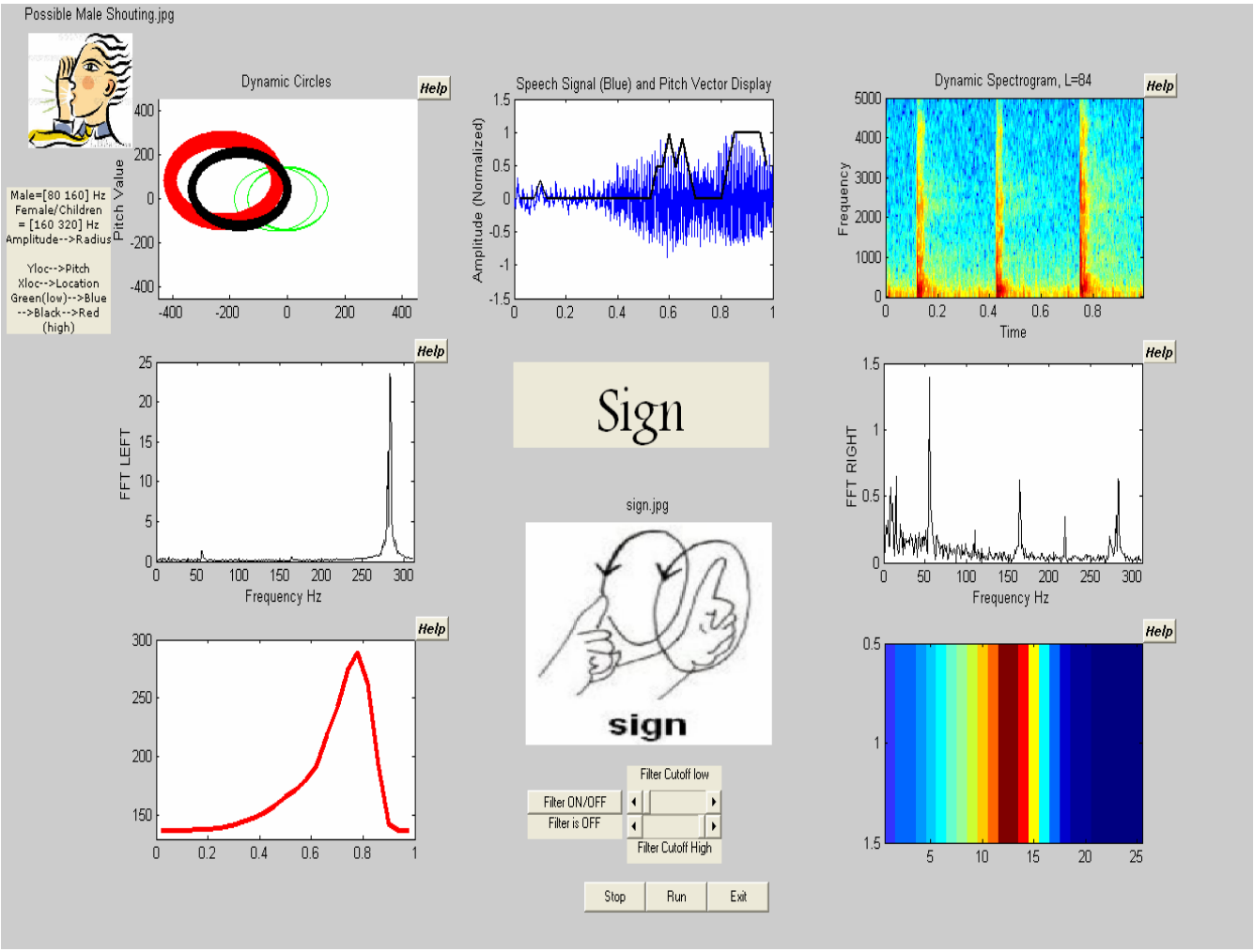

BY: Jimmy Azar

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Abstract:

In this paper, we investigate several means of visualizing both ambient and speech sounds and present a fusion of different visualization displays into one program package that would help provide the hearing impaired with a means to an enhanced awareness of their surroundings. The ideas investigated were implemented in software, and the program was evaluated by means of a survey conducted in a school for the deaf.

Acknowledgments

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Also, we would like to thank the administration and students of the 'Learning Center for the Deaf'-Baabda for their helpful feedback and participation in the survey.

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CHAPTER 1: INTRODUCTION

1.1 PROBLEM DESCRIPTION & OBJECTIVE:

There are approximately 219,480 (5.487%) deaf people in Lebanon [1] and nearly 10,000,000 persons with hearing impairments and close to 1,000,000 who are functionally deaf in the United States. Without Assistive Technologies, there is no possibility for the hearing impaired to recognize sounds efficiently.

Medical or surgical solutions such as cochlear implants may not always be possible. Methods such as mapping frequency and intensity of sound to the frequency and intensity of vibrations (as in an alarm pillow or vibrating movie seat) are limited compared to what visual displays may offer. Other means involving flashing lights remain limited due to their dependence upon some prior knowledge of a characteristic of the specific sound signal expected. The Positional Ripples Display [2] as a proposed method of conveying information of amplitude and location has its limitations in that the physical setup (array of distributed microphones, etc.) may be involved and remains bound to a stationary workplace requiring prior knowledge of the architectural setup. The idea of spectrogram visualization [2-3] is further emphasized in this paper to reveal the capability of such a display in pattern visualization. Furthermore, other sound to image mappings were implemented in addition to speech recognition and Speech-to-ASL translation. By this, we would have provided a comprehensive means of acquiring information from verbal and non-verbal sounds.

CHAPTER 2: REVIEW

Before discussing the Design process, it is essential to present a literature review on existing papers and research that address the problem of non-speech sound visualization for the deaf. This section is intended as a smooth introduction to the topic. It attempts to condense the theory behind the principles used in the visualization process.

Here, we present a review on the different prototypes proposed in [2,3].

2.1 EXISTING TECHNOLOGIES FOR SOUND AWARENESS

We experience many non-speech sounds in every day life such as, mobile phone calls, door knocking, chair movement, typing, etc. These sounds provide us with critical information about events occurring in our surroundings. People make use of these sounds to gain awareness of the state of their environment. For example, at home, the sound of doorbells provides awareness of visitors; also, the sound of a fire alarm is a critical notification of danger.

Without assistive technology, deaf people would not be able to recognize such sounds. Consequently, they must use assistive techniques relying on sensing or flashing lights in order to be aware of these ambient sounds [3].

Many devices and equipments are available to help deaf and hearing impaired people improve communication, adapt to their environment, and function in society more efficiently. For example smoke alarms, phone ringing, and alarm clocks can all be converted to vibrating mechanisms or flashing lights for notification. The figure below

lists the most important existing equipment and means present today. Analysis in terms of cons and pros follows later in Section 3.1.

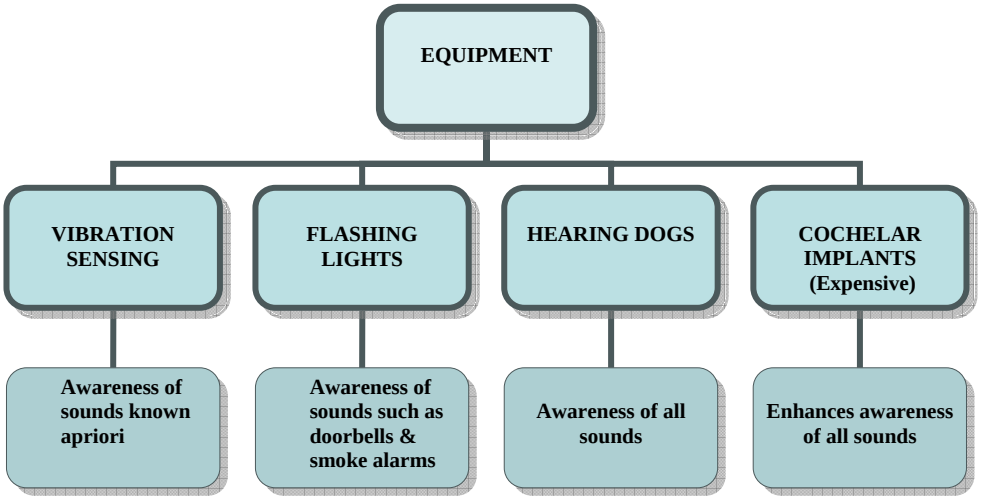


Figure 1: Existing Techniques of Sound Awareness for the Deaf

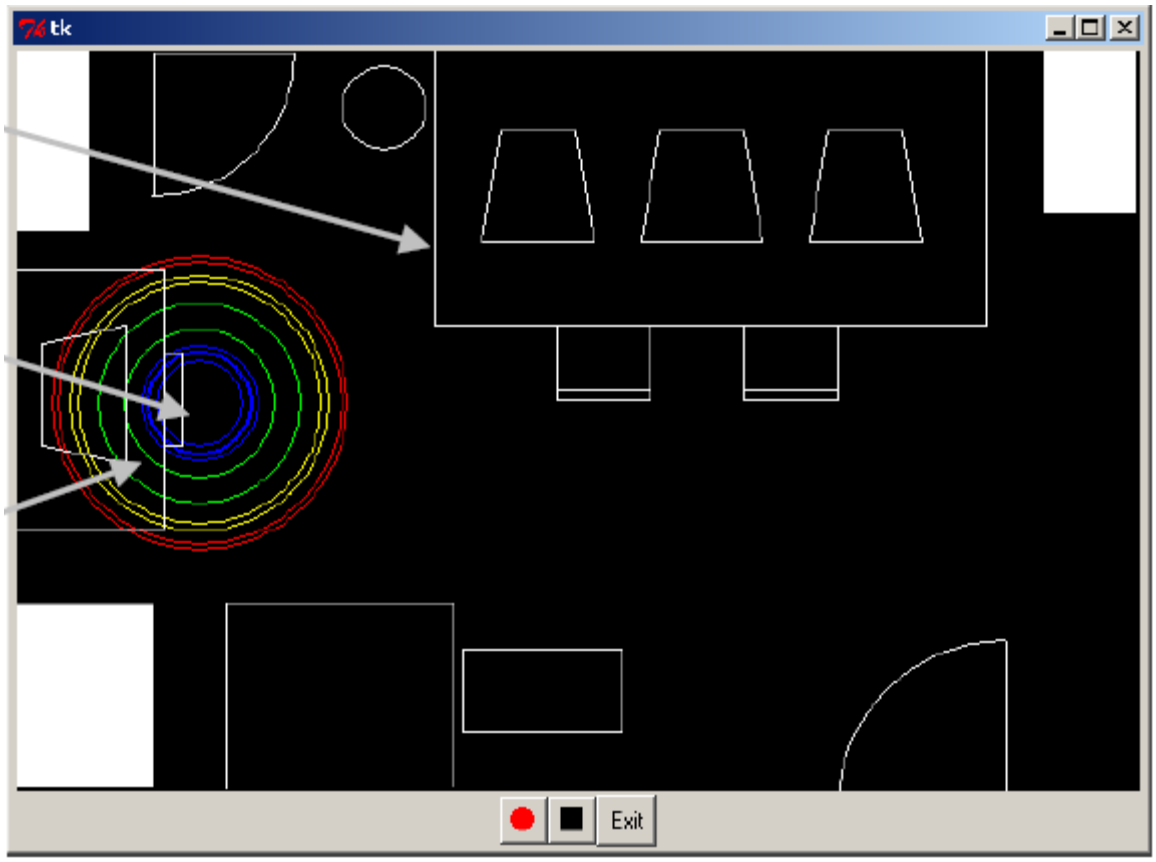
Flashing lights may be used to provide awareness of sounds that inform the person about specific events which require full attention such as fire alarms, door knocking, and phone ringing [9]. All these require a quick response or reflex.

2.2 VISUAL DISPLAYS

Non-speech sound visualization has been proposed in [2, 3, 8]. In [2] the authors explored and analyzed the techniques used by deaf people for sound awareness; they conducted interviews with deaf and hearing impaired participants in the aim of designing an efficient method of sound visualization, and based on these interviews, they concluded that the most important aspect of this visualization is to include the source of sound and

to draw sound as waves (circles). Based on these results, two sound displays have been presented. One is based on a spectrograph and the other is based on positional ripples.

In the spectrograph scheme, height is mapped to pitch and color is mapped to intensity (red, yellow, green, blue, etc.). In the positional ripple prototype, the background displays an overhead map of the room. Sounds are depicted as rings, and the center of the rings denotes the position of the sound source in the room [2]. The size of the rings represents the amplitude of the loudest pitch at a particular point in time. Each ring persists for three seconds before disappearing [2].

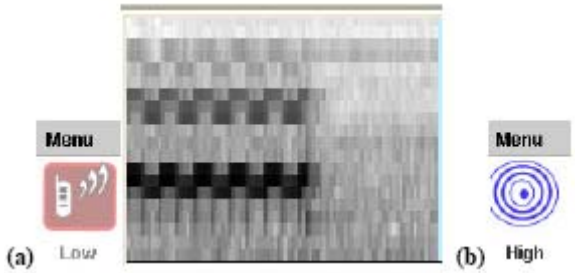


[2] Figure 2: Speech Visualized by Positional Ripples.

This arrangement however is impractical since it requires prior knowledge of the architectural setup (e.g. office); also it is expensive in terms of equipment setup (array of microphones placed at certain corners in the room) and is also not portable (bound to the workplace environment).

In [3], new models have been proposed, based on the previous work done in [2]. The authors proposed two models. The first model, the single icon scheme, displays recognized sounds as icons, located on the upper right corner of the user's computer monitor. It was used throughout the analysis and was shown to give good results. According to the survey performed in [3], all participants liked it because it identified each sound event.

The disadvantage of this method however is the actual need for prior knowledge of the type of sound to be detected; hence the icon scheme in this regard has its limitations. The second model, the spectrograph with icon visualization, improves over the single icon model in that it combines the black and white spectrograph model [2] with the single icon model. The main idea behind this kind of mapping is a desire to associate a particular sound with its shape on the spectrograph [3].



[3] Figure 3: (a) Implemented version of spectrograph with icon showing a phone ringing. (b) Single icon showing a loud recognized sound with low frequency.

CHAPTER 3: DESIGN & ANALYSIS

After thorough study and analysis of the theory behind the existing techniques employed by the deaf, we went on to the design process. This section is mainly concerned with the different design decisions we have made.

First, we will start by presenting a deeper analysis of the existing techniques used by deaf people for sound awareness in order to know where these techniques fell short. Consequently, this critical assessment will contribute a lot to our final design decisions by avoiding the drawbacks of these techniques.

The remainder of this chapter is organized as follows. Section 3.1 presents an analysis of the existing assistive technology for sound awareness. Detailed description on the visual displays proposed is presented in following sections.

3.1 ANALYSIS OF THE ASSISITIVE TECHNOLOGIES

In this section we discuss the advantages and disadvantages of existing assistive means for the hearing impaired. This will help explain why we chose a sound to visual mapping.

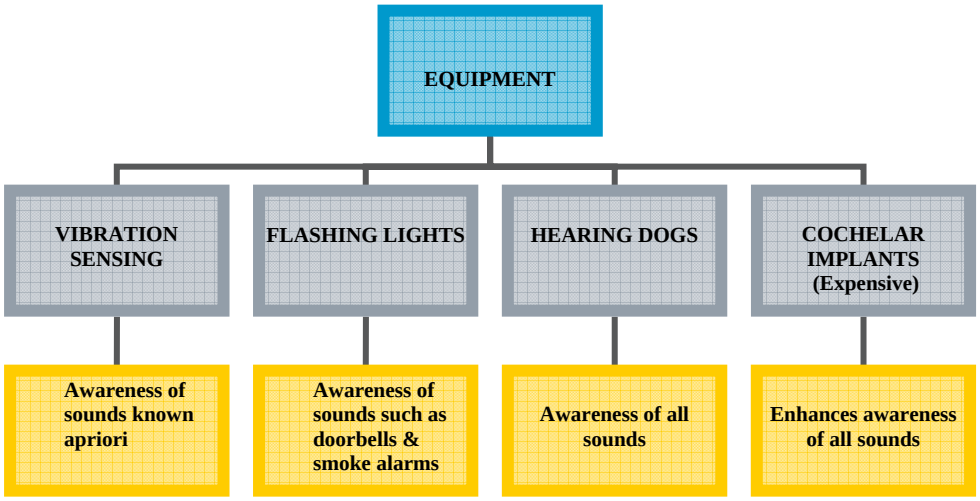


Figure 4: Existing Techniques of Sound Awareness for the Deaf

Assistive technologies and methods such as Phone Flasher, hearing dogs, etc. are impractical in the sense that they require high initial investment, are limited, or need ongoing maintenance. In addition, we realized that all the existing commercial techniques used a separate costly system to notify deaf with sounds, and each system detects a specific kind of sound. For instance, the flashing light technique was used specifically for telephones and doorbells (notification sounds). In addition, other drawbacks stem from the fact that these techniques failed to make the deaf able to depict the state of such appliances; for instance, how can the deaf identify music and the presence of others for example? Hence, many sounds have implicit interpretations that indicate their state. Awareness of such sounds by the existing technologies is limited.

Based on the observations stated above, we decided to develop a software tool that integrates several techniques in addition to a new mapping, and which shows better performance in terms of visualization and detection of speech and non-speech sounds.

The following table summarizes the discussion above:

Technique	Application	Characteristics	Cons
Vibration sensing	-Awareness of sounds that create vibration, e.g. sensing footsteps, feeling that a computer is on	-Does not require focus of attention -Supports some ambient and some notification sounds	-Depends on infrastructure (e.g., having hardwood floors)
Flashing lights	-Awareness of telephones, doorbells (Notification sounds)	-Interrupt based, rather than polling -Supports notification sounds	-Fixed visual attention -Must hook up each device
Hearing Dogs	-Awareness of all sounds	-Interrupt-based -Supports notification sounds only	-Requires ongoing maintenance -Requires a priori training per sound
Visual Inspection	-Multiple applications, e.g., Steam for a kettle, Looking out the window for the arrival of a guest	-Sometimes is the only alternative -Some ambient, some notification	-Different for each sound -Polling rather than interrupt based
Hearing Aids, Cochlear Implants	-Enhancing existing hearing, but not at the fidelity of "normal" hearing	-Enhances awareness of all sounds	-Requires training for interpretation of sounds -Results vary by case

[2] Table 1: Taxonomy of Existing Sound Awareness Techniques for the Deaf

Sight is one of the most important senses. Other senses like touch (vibrating pillow or vibration simulation mapped to intensity while watching a movie) have more limitations when it comes to representing sounds. The technique of flashing lights to indicate certain events is also limited and requires prior installation and knowledge of the type of sound (hence impractical). We are trying to integrate speech and pattern visualization in our program to account for all these needs.

3.2 DESIGN SPECIFICATIONS

We have established the limitations of currently existing awareness techniques for the Deaf. What remains is obtaining the design specs—the target of our design. This is mainly obtained by examining the needs of deaf people. Based on the feedback obtained from the 'Learning School for the Deaf'—Baabda, the needs for deaf people in general may be summarized as:

i) The need to understand Spoken Speech when present

ii) The need to be aware of certain events that tend to recur such as phone ringing and door knocking.

iii) The need to be aware of the source of sound (direction)

In addition to these, we have added

iv) The ability to observe the variation in the loudness of sound (increasing/decreasing/ monotone)

v) The ability to distinguish the different frequency components of sound (low pitch/high pitch)

vi) The ability to identify whether the sound belongs to a male/female ,

shouting/speaking, or Nonhuman Loud, or Telephone ringing.

We have tested the success in achieving these specs by means of a survey conducted in a school for the deaf. The results are summarized in Section 5 (Evaluation).

In designing for these specs, we had to acquire two input sound signals to localize sound, calculate root mean square values to tag loudness levels, precondition the signals, and apply a pitch extraction algorithm, DFT analysis, and spectrogram setup. The detailed implementation of these ideas is next examined in section 4.

3.3 DESIGN CONSTRAINTS

The program to be designed had to be flexible and implementable onto any usual microprocessor. The constraints on the program itself can be summarized as:

i) Should provide a user friendly Graphic User Interface.

ii) Should support a reasonable vocabulary size for speech recognition

iii) Should yield accurate detection of sounds, their location, and loudness, etc.

iv) Should be able to implement a fusion of Speech and NonSpeech sound Visulaization onto 1 Main Window.

CHAPTER 4: IMPLEMETATION

4.1 INTRODUCTION

4.1.1 DESIGN OVERVIEW

Several sub-windows were designed with each providing specific sound information. All these sub-windows are dynamically changing as sound is continuously inputted (live), and they are all simultaneously displayed within a Main Window. The sub-windows may be divided into those visualizing information of ambient sounds and those displaying speech. Belonging to the first category are the Dynamic Circles Display (for visualizing pitch, sound location, and loudness), Dynamic Spectrogram Display (for pattern visualization), Pitch Vector Display (for speech visualization and emphasis of pitch presence/absence), RMS Line Display, and RMS Image Display (for visualizing changes in sound intensity via a plot or color strip image). Belonging to the second category are the Speech Recognition Display and the Speech-to-ASL Display.

4.1.2 SYSTEM SETUP

The program was written using MATLAB version 6.1 and run on a 1.7 GHz Intel Pentium processor. The data acquisition system consisted simply of a pair of identical microphones introduced into the 'Line in' socket. The system was adjusted for Line in Recording. Sound signals were inputted via two separate channels each corresponding to a microphone. Hence, the input consists of two separate sound signals which may be processed separately in the case of FFT Left/Right displays as wells as in the Dynamic Circles display, but are combined to present displays for the other sub-windows.

Section 2 of the paper discusses the various means of visualizing ambient sounds. Different displays are proposed and implemented.

Section 3 discusses speech visualization which includes the implementation of speech recognition and speech to ASL translation followed by an overall view of the entire Main Window Display.

Section 4 includes a summary of the statistical results gathered from a survey conducted in a school for the hearing impaired in the intention of assessing the usefulness and performance of our program. Possible applications are also discussed.

Section 5 contains the conclusion of our work and suggestions for possible future improvements.

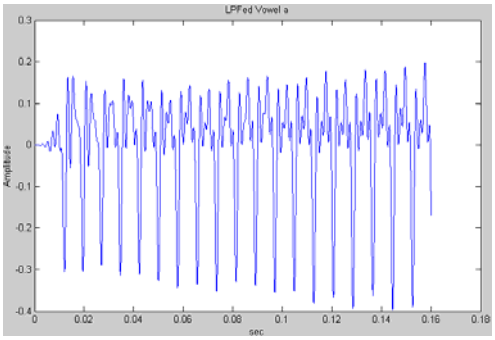
4.2 AMBIENT SOUND VISUALIZATION

4.2.1 PITCH EXTRACTION ALGORITHM

The Autocorrelation Function (ACF) as defined in equation (1)

$$\mathbf{R_y[k] = < y[n]\, y[n-1] >} \qquad \textbf{(1)}$$

(where **<.>** corresponds to the mean) is essential for extracting the pitch value of a sound segment. Prior to extracting the pitch, preconditioning is applied mainly in the form of Center Clipping and median filtering. Pitch contributes to significant peaks in the Autocorrelation function (ACF), however in between there may exist secondary peaks corresponding to the vocal tract transfer function. These peaks may cause errors in the automated pitch detection process. Center Clipping the speech signal by considering only the peaks that overpass 68% of the minimum or maximum values of the signal eliminates these undesirable peaks and hence allows for a more accurate detection of pitch [4].



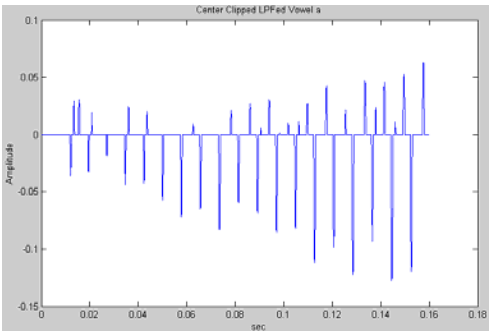


Figure 5: (a) LPF'ed Vowel Segment('a'); (b) Effect of Center Clipping

Median Filtering is then applied over the speech segment before computing the ACF in order to remove outlier components outside the human voice range. The sound segment is then divided into intervals of 50 milliseconds overlapping midway. To avoid unnecessary peaks, the segments are further Low Pass filtered using a finite impulse response (FIR) filter of order 125 with a cutoff frequency equal to 500 Hz since the upper limit of the human pitch range is ~320 Hz. Finally the ACF is processed for each sub-segment to yield a pitch value for each sub-segment.

Pitch may be extracted from the ACF by means of a simple algorithm described below:

(i) The positive part of the symmetric autocorrelation function (i.e. from Ry(0) →end) is considered.

(ii) Identify/Find the index at which the first peak (that of Ry(0)) actually ends.

(iii) Search for the 2nd peak 'pk' which is just the maximum value of the remaining part of the function starting from the discovered index in (ii).

(iv) The correct index of the second peak is then converted to real time by multiplying with the sampling period Ts to obtain 'pk_t'.

(v) To be considered a valid peak, the second peak pk has to be greater than 30% of the first peak Ry(0).

The pitch is then:

$$\begin{cases} \dfrac{1}{pk_t}, & if \;\; pk > 0.3\, Ry(0) \\ 0, & Otherwise \end{cases}$$

For instance pitch is equal to 0 for consonant 'sh' which is of an unvoiced fricative nature as compared to a stressed vowel segment like 'a'.

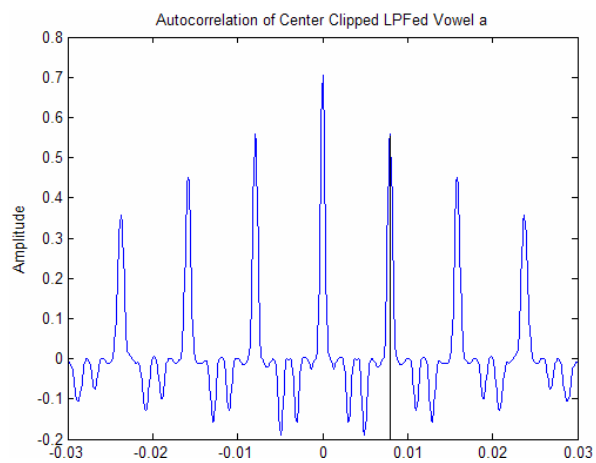


Figure 6. ACF of Center Clipped Vowel Segment

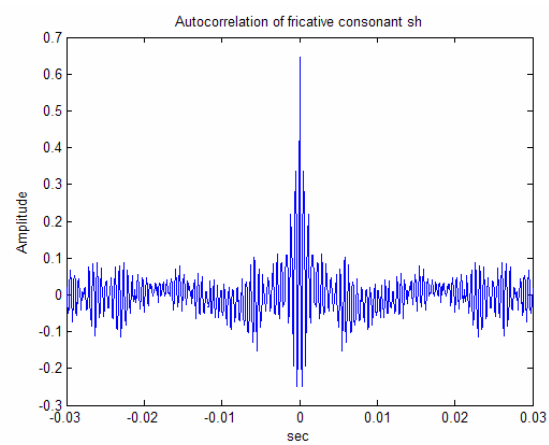


Figure 7. ACF of consonant 'sh'

4.2.2 PITCH VECTOR DISPLAY

The presence of definitive pitch is often indicative of intelligible sounds. For example, pitch is apparent in speech mainly due to vowels and sound originating from the vocal cord (e.g. 'zzz') as well as in musical instruments which possess a large range of pitch values. Fricative or consonant unvoiced sounds which in the case of speech usually result from teeth, tongue, and lips (e.g. 'sss') do not possess a pitch value. An incoming sound segment is further divided into 50 msec intervals overlapping midway, and the 'detect_pitch' function is used to obtain the corresponding pitch value for each of these sub-segments. The Pitch Vector Display is basically a plot of these values after normalization. The normalized pitch values are plotted (in black) on top of the speech signal segment (in blue) in a sub-window in between the Dynamic Circles and Dynamic Spectrogram sub-windows. The idea is to display the presence of pitch and its duration, but more importantly to give a clearer view of the black pitch line as indicative of pitch activity. This sub-window is to be considered complimentary to the two other displays in that it further emphasizes the presence or absence of pitch. The Pitch Vector is displayed coincided with a plot of the acquired normalized speech segment with the dc mean removed (Figure 8a).

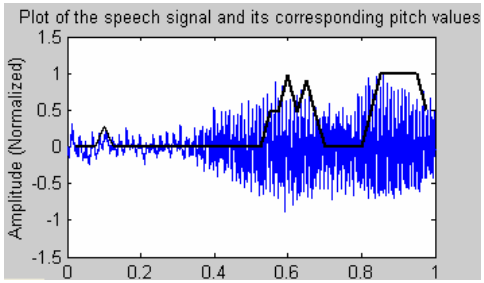


Figure 8a. Pitch Vector Display (black line) with normalized sound segment overlapping

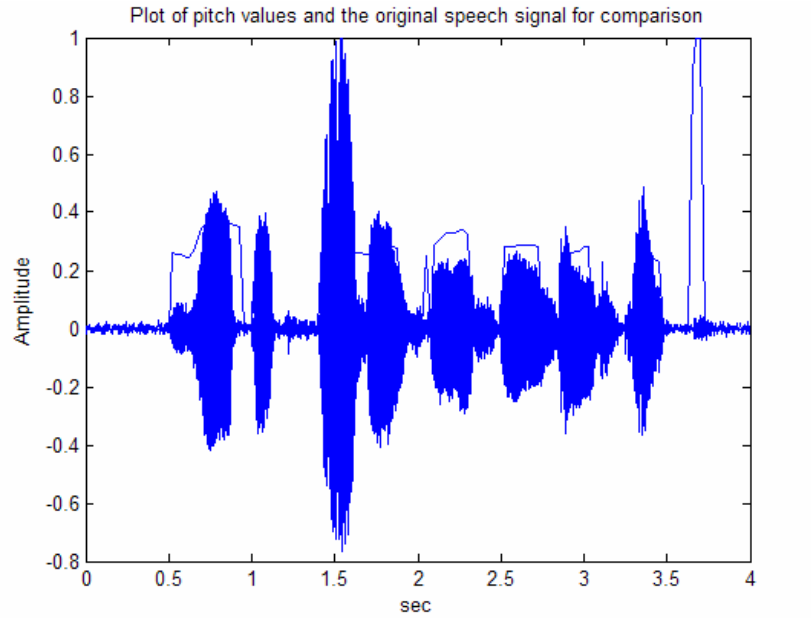


Figure 8b. Plot of Pitch values and original speech signal (compare/match)

4.2.3 DYNAMIC CIRCLES DISPLAYS

When asked to represent sound, ripples or circles are usually adopted [2]. Hence, we developed The Dynamic Circles Display which combines sound localization and pitch extraction. The location of the center of the circle is indicative of the location of sound linearly between the two microphones and the mean pitch extracted from the acquired sound segment. The physical location between left and right is mapped to the x-location of the center, and the mean pitch value is mapped to the y-location. The pitch vector discussed often contains several 0 elements which are then removed and the remaining values averaged to yield a mean pitch value for the entire segment—this is the y-ordinate of the center of the circle. By this, non-pitch segments are ignored and the mean pitch corresponds to that of the pitch-possessive segments. If the vector is entirely a zero vector, then the mean pitch value is 0. The loudness of the sound segment is mapped

simultaneously to the radius, color, and thickness of the circle. A new circle is displayed every second with every incoming sound segment. The window is cleared every 5 seconds so as to display a temporary history of 5 circles. In the case of speech, the range in which the value falls is indicative of the gender and/or age of the speaker as males have pitch values in the range [80 160] Hz while females and children in the range [160 320] Hz. The display of the loudness of sound may be indicative of the emotional state of the speaker (whether angry, calm, etc).

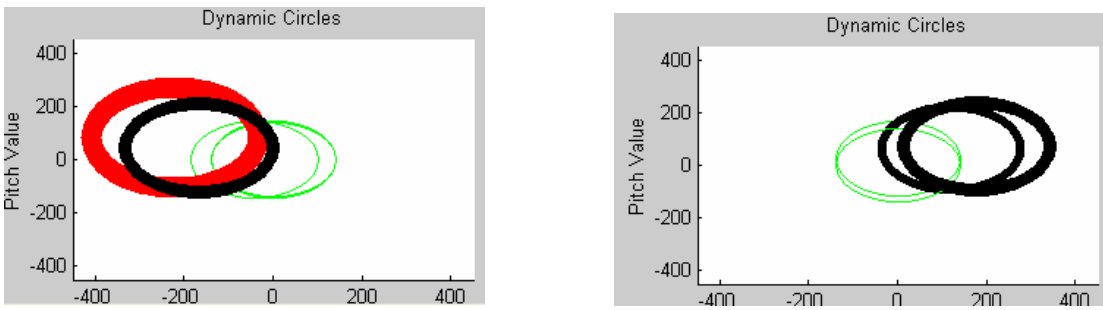


Figure 9. (a) Sound source moving towards left side and becoming louder; (b)Sound source moving towards right side and becoming louder. Pitch values for both is in the range [0 200]Hz. Color Range: Green(low)→Blue→Black→Red(high)

4.2.4 DYNAMIC SPECTROGRAM

A spectrogram is a very useful visual tool for displaying the intensity of frequency components over time segments via color. With sound being highly non-stationary, the Short Time Fourier Transform (STFT) was used in this regard. This display reveals dominant frequency components (mainly formant band striations due to resonance of the vocal tract), their harmonics, and a sense of how the intensity of such components is changing with time. If well trained, words may be read from spectrograms based on the patterns that appear. This however is a difficult task and may be compared to learning a second language. The intention behind the spectrogram display is pattern

visualization of events that tend to recur. Examples of such events are phone ringing, door knocking, and other ambient or peripheral sounds. Thus this is a form of learning by induction and observation—drawing a link between an occurring event and its appearing visual pattern.

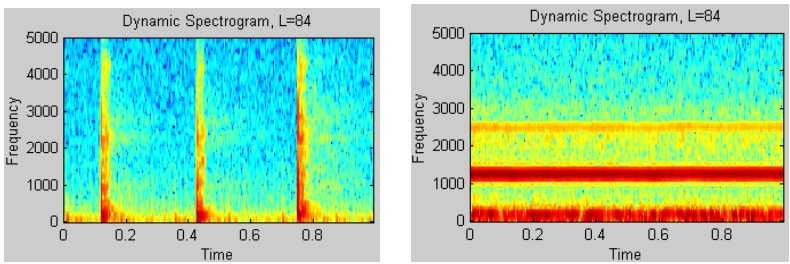


Figure 10. (a) 3 Door Knocks; (b) Someone Whistling

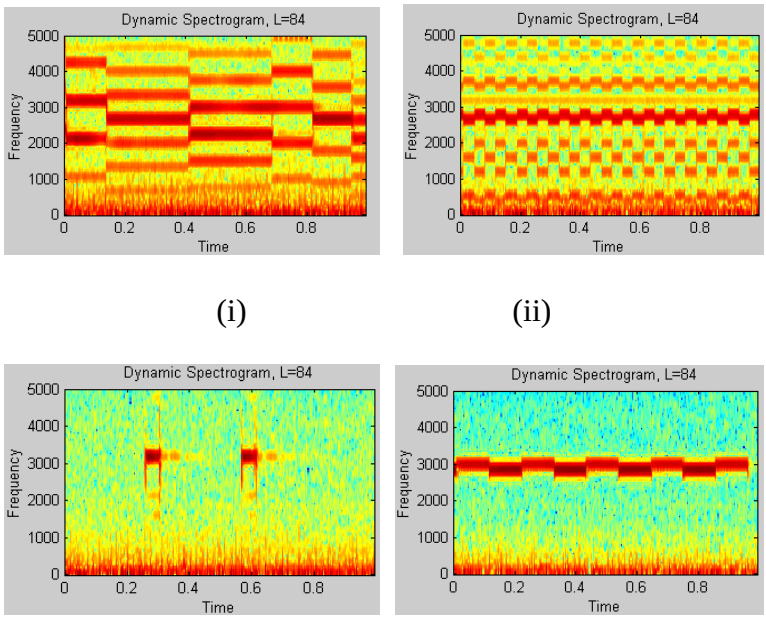


Figure 11. Visual Patterns for Phone Ring Tones (Nokia Model 8850)
(i) 'Nokia' Tone; (ii) Traditional 'low' tone;
(iii) 'Knock-Knock' Tone; (iv) 'Mosquito' tone

The spectrogram used is wide-band with a frequency resolution of 300 Hz. The window utilized within the spectrogram is a Hanning window [5] described by equation (2)

$$\mathbf{w(n) = \frac{1}{2}\left[1 + \cos\left(\frac{2\,x\,\pi\,x\,n}{2M+1}\right)\right], \quad -M \le n \le M} \qquad \textbf{(2)}$$

It should be noted that the tradeoff between time and frequency resolution is described by the uncertainty principle [6] mainly stating:

If an impulse response h(t) treated as a probability density function (PDF) decays to zero faster than $\frac{1}{\sqrt{|t|}}$ for large t, then

$$\mathbf{(\Delta T)\ (\Delta\omega) \ge 2} \qquad \textbf{(3)}$$

Where **(**Δ**T)**, the time duration, is defined as twice the standard deviation σ_h from its zero centered mean μ_h:

$$\mathbf{(\Delta T)^2 = 4\,\sigma_h{}^2 = 4\,\frac{\int_{-\infty}^{\infty} t^2\, h(t)^2 dt}{\int_{-\infty}^{\infty} h(t)^2 dt}} \qquad \textbf{(4)}$$

and **(**$\Delta\omega$**),** the bandwidth of **H(**ω**),** is defined as twice the standard deviation σ_H of the PDF $|H(\omega)|^2$.

$$\mathbf{(\Delta\omega)^2 = 4\,\sigma_H{}^2 = 4\,\frac{\int_{-\infty}^{\infty} \omega^2\, |H(\omega)|^2 d\omega}{\int_{-\infty}^{\infty} |H(\omega)|^2 d\omega}} \qquad \textbf{(5)}$$

4.2.5 LEFT/RIGHT DISPLAY

The input consists always of two sound segments acquired from the left and right microphones. The Discrete Fourier Transform (DFT) is applied on each of the two segments using the Fast Fourier Transform (FFT) algorithm. The magnitude of the DFT is then plotted versus frequency for the left and right signals separately. This display allows visually detecting the dominant and non-dominant frequency components present and comparing these between left and right sides.

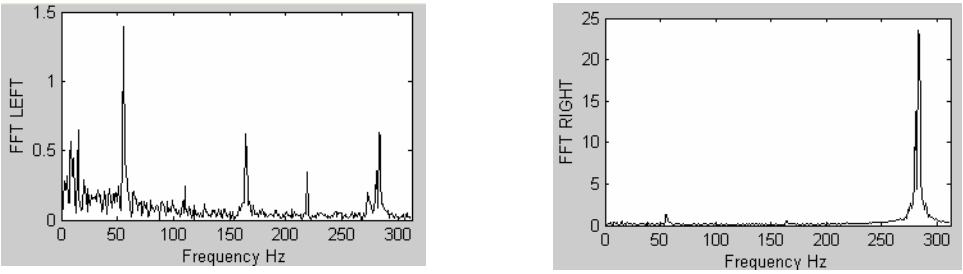


Figure 12. Singing a high pitch 'aaa' from right side
(a) DFT Display (left mic); (b) DFT Display (right mic) shows dominant frequency component in range [250 300] Hz.

4.2.6 RMS LINE DISPLAY

The root mean square value of a sound segment is a direct indication of loudness. The speech segment is divided into sub-intervals, and the root mean square value is calculated for each sub-interval. The result is then plotted to reveal how loudness is varying. This is of course complimentary to the thickness/color/and radius of the dynamic circle display with the difference that the RMS display has a faster, clearer, and more accurate response to changes in loudness levels due to the division of sound into sub-intervals.

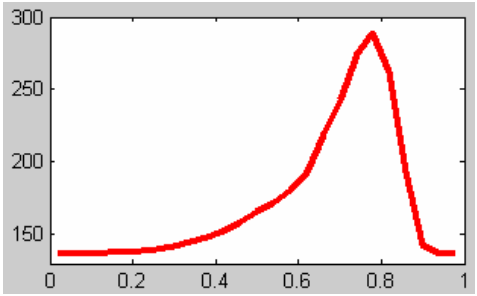


Figure 13. 'EEEE' sound showing exponentially rising loudness intensity.

4.2.7 RMS IMAGE DISPLAY

This display is an alternative to the RMS Line Display. Instead of plotting the rms values versus time, each value is assigned a color strip. The result is an image consisting of consecutive vertical color strips corresponding to consecutively varying loudness levels.

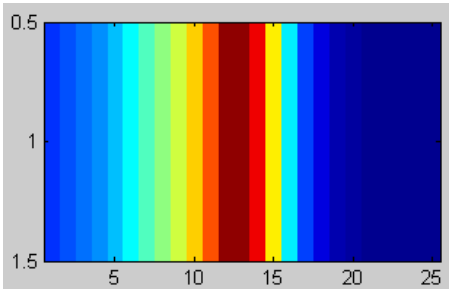


Figure 14. Loudness Intensity Rising then Falling

4.3 SPEECH VISUALIZATION

4.3.1 SPEECH RECOGNITION

In case the sound is speech, recognizing speech is of great significance to the hearing impaired. The vocabulary set could be of any size and is dependent on the training. We have used a set composed of 12 words to complete a few sentences with an additional 'Unknown?' word that is displayed whenever a preset threshold is exceeded indicating that either the sound is far from being speech or is not close enough to any of the existing 12 words. Standard preconditioning steps, including pre-emphasis, were performed, and the Cepstrum method was used in identifying word segments. This is displayed in a central sub-window (Figure 15). The reliability of the Cepstrum method was tested using the Cloning Algorithm [7] (equivalently the more recent Boosting method). The Cloning Algorithm was applied on the training data (cepstral vectors) for 10 iterations to yield a modified average cepstral vector for each word.

Figure 15. Speech Recognition Display of the word 'Sign'

4.3.2 SPEECH TO ASL TRANSLATION

In a deaf community, some people prefer communicating in sign language such as American Sign Language (ASL) which is common is the United States and Canada. Moreover, people who are initially born deaf or with severe hearing impairments face more difficulties acquiring formal written English language in school as they do not hear

it to incidentally acquire it with ease. Hence, it is important to include speech to ASL or Signed English translation within the entire program package. The method of implementation is similar to that of standard speech recognition with the exception that instead of displaying words upon successful matching, static sign pictures are displayed consecutively in a concatenated frame-like fashion. Each of the 13 words has its own display image.

Figure 16. Sign Language Display

4.3.3 SINGLE ICON DISPLAY

The idea of Single Icon Displays [3] was also implemented. Generally, these displays are limited in that they require prior knowledge of the signal expected, but may be quite useful for certain basic events or if the conditions behind a metric match is kept general. For instance certain visual patterns appearing on the spectrogram may be recognized to cause an icon to pop up indicating 'phone ringing' or 'door knocking', etc. For more relaxed conditions, if the mean pitch value falls in a specific region of the human pitch range and according to the loudness/intensity level, an icon may display such interpreted information accurately. We have successfully implemented 7 icons demonstrated as follows:

'Possible Male Speaking'	
'Possible Female Speaking'	
'Possible Male Shouting'	
'Possible Female Shouting'	
'Possibly Non-Human Loud'	
'Possible Telephone Ringing'	
'Question'	

Table 2: Icons Display

4.3.4 GUI CONTROLS

An optional FIR Bandpass filter is also included in the Graphic User Interface (GUI) to allow the user to set lower and upper cutoff frequencies as desired. This would of course affect the display of the sub-windows eliminating any unwanted frequency regions in the spectrogram for instance. Other GUI controls involve stopping (freezing), running, and exiting the program at any time. A Help button for almost every sub-window may be pressed and depressed to reveal a brief explanation of the purpose of that respective window. For example, the help button for the dynamic Circles sub-window has been pressed to reveal the illustrative side-text to the left (Figure 18).

Figure 17. GUI Controls (BP Filter, Stop, Run, Exit)

Overall Main Window Display

The Main Window includes all the sub-windows discussed. All these sub-windows are dynamically changing with the continuously incoming sound signal.

The Main Window Display is shown in Figure 18.

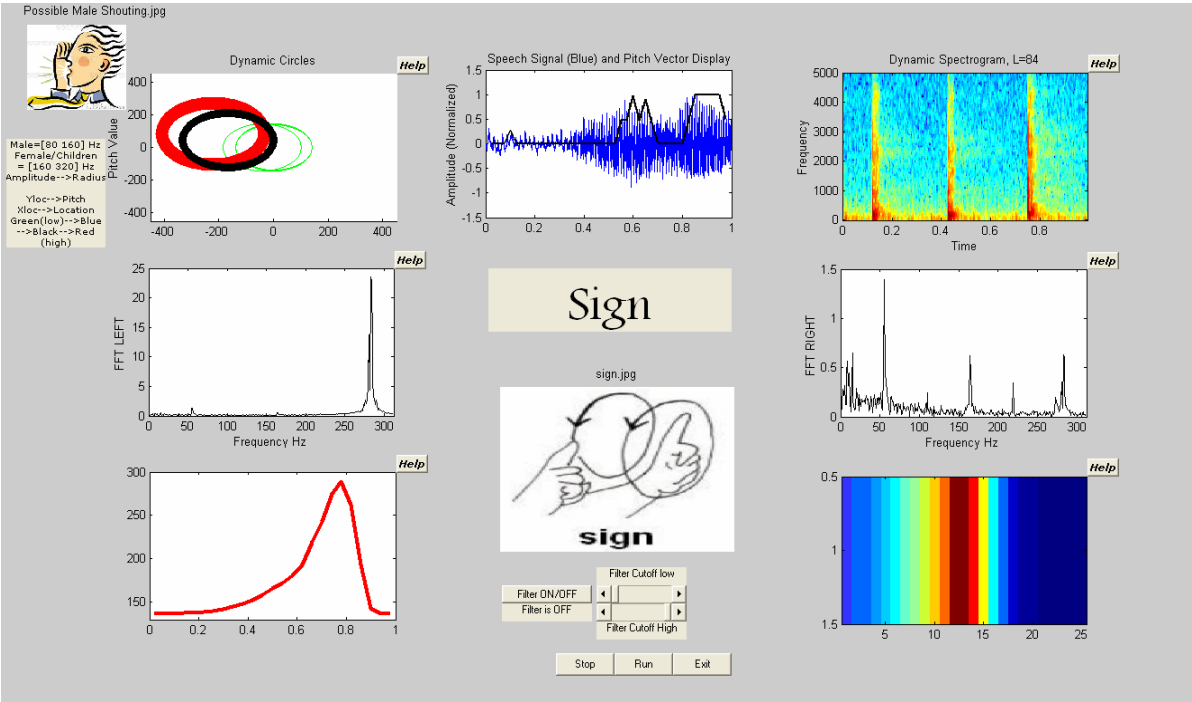


Figure 18. Main Window Display

4.3.5 APPLICATION

The idea behind the program developed is to provide the hearing impaired with more awareness of the environment around them whether in the workplace or at home. The addition/installation of such software onto any computer or PDA for instance would help reduce communication impediments that are usually faced without disrupting other tasks being performed by the viewer. The entire main window of the program would be

displayed for example in a separate corner of the screen while the user would be working in another program. The software could also be integrated into any portable computer based device that would include two built-in microphones or an array of microphones situated on the lower backside of the screen. A small separate device with a reasonable microprocessor supporting the program, sensitive microphones, and a miniature screen may be even used as a portable wrist bound item. The program would continuously provide information of sound source location, loudness, pitch, speech translation, etc.

CHAPTER 5: EVALUATION

The Program was evaluated by means of a survey conducted in a school for the hearing impaired. The 10 participants had various degrees of hearing impairment from partial deafness to total deafness. The survey focused on assessing the significance and usefulness of the entire program as a unit as well as comparative assessment among the different sub-windows. A summary of the results of the survey is depicted below in Figures 19 and 20.

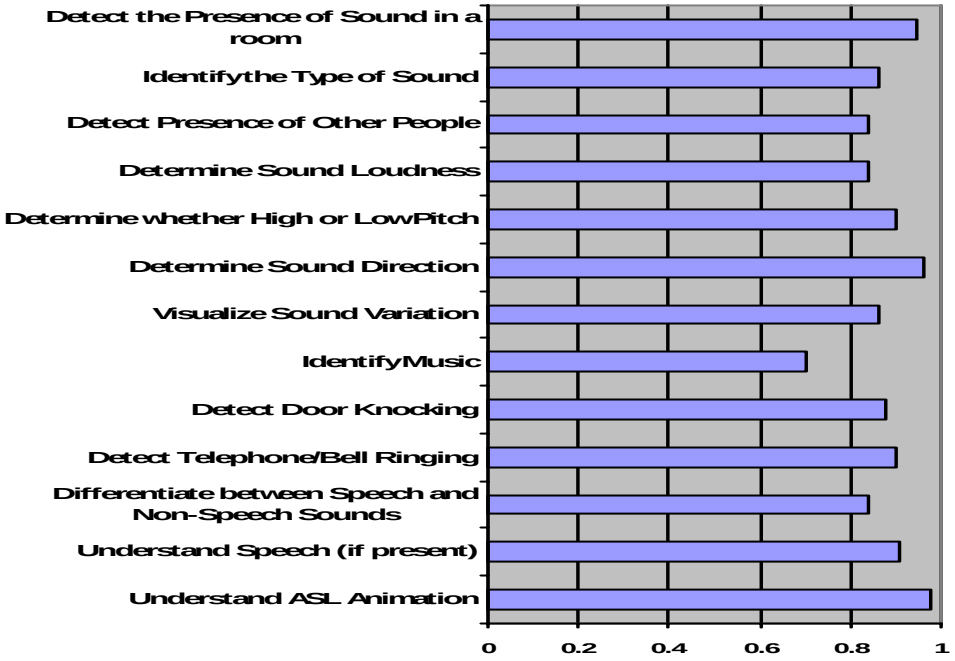


Figure 19. Assessment of Functionality

Figure 19 is based on the participants' assessment of the degree of success of the program in achieving the targets specified.

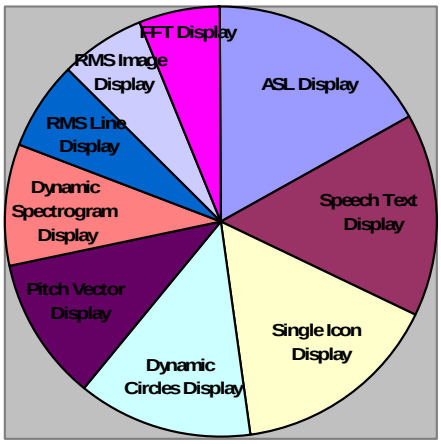


Figure 20. Relative Significance of the Different Displays

The comparative assessment of the various sub-windows (Figure 20) showed that the participants found that the ASL Display was the most useful followed by Speech Text Display, and that the least useful sub-widow was the FFT Right/Left Display.

The overall usefulness of the entire program prototype was given an average of 88% grade by the participants. The participants were in general excited by the output of the program and showed great interest in the possibility of having this implemented onto a small portable device or cell phone.

CHAPTER 6: CONCLUSIONS & FUTURE WORK

The program discussed in this paper would provide the hearing impaired with varying types of information extracted from sound. The program was written in the simulation environment of MATLAB. Future work would include actual hardware implementation in more efficient program languages such as C before optimization into Assembly Code. Speech Recognition may be improved and transformed into a Continuous Speaker-Independent system based on phoneme recognition and HMM. Furthermore, the program may be developed even further to include two-way communication means. For instance, the hardware could be designed to support a two-sided screen; the screen facing the user would show the program display and the screen opposing the user would be used to display any text typed by the user (which simultaneously shows on the screen facing the user). Moreover, with the advances in ASL-to-Speech recognition software, a computer camera may be installed to provide a further option of signing instead of typing text on a keyboard. The sub-windows may be designed with the option of collapsing any frame whenever the user feels it is distracting and/or not required for the moment.

REFERENCES

[1] Ethnologue report for Lebanon http://www.ethnologue.com/show_country.asp?name=LB

[2] F. Wai-ling Ho-Ching, Jennifer Mankoff, James A. Landay, (2003). "From Data to Display: the Design and Evaluation of a Peripheral Sound Display for the Deaf", In Proceedings of CHI 2003. 8 pages.

[3] Tarra Matthews, Janette Fong, Jennifer Mankoff, "Visualizing Non-Speech Sounds for the Deaf", In Proceedings of ACM SIGACCESS conference on Computers and Accessibility (ASSETS). Baltimore, MD, pp. 52-59, 2005.

[4] L.R. Rabiner and R.W. Schafer, "Digital Processing of Speech Signals", Prentice-Hall, 1978, pp. 150-154.

[5] Sanjit K. Mitra, "Digital Signal Processing A Computer-Based Approach", 3rd ed, McGraw-Hill, pp.533

[6] David Slepian, "Some Comments on Fourier analysis, uncertainty and modeling", SIAM Review, Vol. 25, No.3, pp. 379-393.

[7] Mohamad Adnan Al-Alaoui, Rodolphe Mouci, Mohammad M. Mansour, and Rony Ferzli, "A Cloning Approach to Classifier Training", IEEE Transactions on Systems, Man, and Cybernetics—Part A: Systems and Humans, Vol. 32, NO. 6, November 2002, pp. 746-752.

[8] "Sound visualization: Sound to animation" 2001. <http://www.mat.ucsb.edu/~woony/research/fall01/mat596b/> Accessed 2005 Dec 15.

[9] American Speech-Language-Hearing Association."Assistive Technology. What are Assistive Listening Devices? " <http://www.asha.org/public/hearing/treatment/assist_tech.htm> Accessed 2005 Dec 2.

APPENDIX (SURVEY QUESTIONS)

The following Questionnaire is intended after the entire program is complete in the Spring Semester. It will be addressed to a Deaf Community (such as a School for the Deaf). The survey will help assess the degree of success of the program in providing an integrated tool for visualizing sound.
The questions are still preliminary subject to change or additions depending on the final state and features of the sub-programs yet to be designed.

Questionnaire

Topic: The purpose of this survey is to assess the design of a system that visualizes speech & non-speech sounds to help people with hearing impairment. We appreciate your support by responding to the following questions.

Section I. Tell us about you.

1) What is your full name? (Optional)

2) Describe your occupation?

3) Do you have any hearing impairments? If so, please describe it and tell us when you started experiencing hearing loss?

4) Do you have any non-correctable visual impairment? If you have normal vision with glasses or contact lenses it does not count.

Section II. Work

1) Describe your working environment (office, cubicle, office building, etc.)

2) In an average working day, how many hours do you spend at work?

3) Describe all the sounds that you hear in a typical day at work both inside and outside the office.

4) What sounds help you do your job (telephone, customers, co-workers)?

5) What sounds do you listen for (footsteps into office, keyboard clicking, boss)?

6) What sounds distract you from your work?

Section III. Home

1) Describe your home environment (apartment, house, commercial area).

2) In an average day, how many hours do you spend at home? _____

3) What sounds do you normally encounter at home (people, barking, phone, cars)?

4) What sounds do you or would you attentively listen to (phone, doorbell, etc.)?

5) What sounds do you find distracting or annoying at home?

Section IV

The purpose of this survey is to aid in the design of a computer program to help the hearing disabled by visualizing and monitoring speech and non-speech sounds, thereby allowing one sense (sight) to process information meant for a weaker or non-existent sense (hearing).

The following questions will help assess and determine the design of the final device.

On a scale of 1-7 (1=Least, 7=Most) please rank to what degree did the program allow you to detect each of the following, and please explain briefly why.

1) Detect the presence of sound in a room

1 2 3 4 5 6 7

2) Detect both the presence of sound and identify the type of sound in or near a room.

1 2 3 4 5 6 7

3) Detect the presence of other people in a room.

1 2 3 4 5 6 7

4) Determine the volume or intensity of sound in a room

1 2 3 4 5 6 7

5) Visualize whether the sound in a room is of high pitch or low pitch.

1 2 3 4 5 6 7

6) Determine what direction the sound is coming from

1 2 3 4 5 6 7

7) Having a sense of the way the sound changes over time

1 2 3 4 5 6 7

8) Being able to detect if someone is playing music

1 2 3 4 5 6 7

9) Being able to detect if someone is knocking on the door or if a doorbell is ringing

1 2 3 4 5 6 7

10) Being able to detect if the telephone is ringing

1 2 3 4 5 6 7

11) Differentiate between speech and non-speech sounds.

1 2 3 4 5 6 7

12) Understand speech (if present)

1 2 3 4 5 6 7

13) Understand ASL animation

1 2 3 4 5 6 7

Please answer the following Questions?

14) Which of the windows did you find most useful? And Why?

15) Which of the windows did you find least useful? And Why?

16) Circle Languages you are familiar with (know well)?

a) Written English b) Signed English

17) Which did you prefer in the program? (circle only one)

a) Reading Speech b) Viewing Signed English

18) How would you rank the usefulness of the windows in descending order (most useful to least useful)—type in the letters below.

(a) Dynamic Spectrogram Display
(b) Dynamic Circles Display
(c) FFT Left/Right Displays
(d) RMS Line Display
(e) RMS Image Display
(f) Pitch Vector Display
(g) Speech Text Display
(h) Speech-to-ASL Display
(e) Single Icon Display

19) How would you rank the Usefulness of the entire program as one unit? (1=Not Useful, 7=Very Useful)

1 2 3 4 5 6 7